LITERATURE REVIEW

Our Project: Uncertainty-Aware Contextual Recognition of Sign Language Sequences
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Paper Selected:
In the paper presented by Koller, Zargaran, Ney, and Bowden [1], the authors describe a hybrid approach, called Deep Sign, which uses a combination of a Convolutional Neural Network (CNN) and a Hidden Markov Model (HMM) for the task of Continuous Sign Language Recognition (CSLR). The selected paper is highly relevant to our final project, as it describes how deep visual models and probabilistic temporal models can be used to address a major problem in sign language recognition.

Paper summary:
The paper's main idea is to embed a CNN into an HMM-based recognition pipeline and interpret the CNN's outputs in a Bayesian way. The authors emphasize that sign language data is inherently sequential, and systems that ignore sequence structure during training or evaluation often perform poorly on realistic CSLR tasks. Their hybrid design uses the CNN for strong discriminative visual modelling while relying on HMMs for sequence modelling and decoding.

In the proposed framework, an input video is treated as a sequence of frames, and the goal is to find the most likely sequence of sign words (glosses) that explains the observed visual input. The authors describe the standard decomposition used in statistical recognition systems: a visual model and a language/sequence model. To handle temporal variation, they use an HMM where each sign-word is represented by a set of hidden states with left-to-right transitions. Sequence decoding identifies the best-matching sign sequence under transition constraints and observation probabilities.

A key technical point is how the CNN outputs are used. Neural networks naturally provide posterior probabilities, while an HMM typically needs likelihoods. The paper explains how to convert posteriors to likelihoods using Bayes' rule and state priors, allowing the CNN to serve as the HMM's observation model in a principled way. This is central to why the approach is described as a hybrid CNN–HMM system trained end-to-end within the recognition framework.

The authors evaluate the approach on three benchmark CSLR datasets and report substantial performance improvements measured by Word Error Rate (WER), a sequence-level metric. They also compare their hybrid method to a tandem approach, in which a CNN extracts features and then models them by another HMM system, and find that their hybrid method performs better. The overall conclusion of this paper is that a combination of the discriminative power of CNNs and the sequence modelling ability of HMMs can, in fact, perform better than many techniques that use a sliding window, temporal normalization, or evaluation techniques that do not strictly follow sequences.

Implications of this paper to our project:
Our final project aims to build an uncertainty- and context-aware sign recognition system and compare a frame-based baseline with a sequence-aware model that leverages temporal context to improve predictions. The Koller et al. paper is directly applicable in three major ways:

1. **Justification for sequence modelling**
   Koller et al. clearly motivate why sign recognition should be treated as sequence decoding rather than independent frame classification. This supports the core framing of our project: that frame-level classifiers can be improved by modelling temporal structure.
2. **Concrete architecture inspiration**
   Our planned system similarly uses a CNN to produce per-frame or per-segment probability distributions over classes. Then it applies an HMM-based decoder to produce a globally

consistent output sequence. The paper provides the conceptual and methodological basis for integrating the components in a principled Bayesian fashion.

3. **Evaluation at the sequence level**
   The paper focuses on sequence-level evaluation (WER) rather than per-frame accuracy, which is not reported, aligning with our plan for evaluating improvements in sequence prediction. Although we might use different metrics in the vocabulary setting, the overall principle of evaluating the system as a sequence predictor is immediately applicable.

Our approach:
Although the paper strongly informs our approach, our project will not reproduce their end-to-end CSLR benchmark system. Instead, we will adapt the key ideas to a course-feasible and more interpretable setting:

- **Scope and vocabulary**
  Koller et al. target large, real-world CSLR datasets and evaluate on complex corpora. Our project will focus on a constrained vocabulary (e.g., ASL alphabet or a limited sign set) to ensure reproducible training and clear evaluation.
- **Controlled study of uncertainty and context**
  While Koller et al.'s work on CNN output can be viewed as probabilistic in nature, the focus seems more on benchmarking. The primary contribution of our project would be to demonstrate the change in uncertainty with the addition of context, i.e., top-k stability, confidence changes, and before/after decoding comparisons, to explain why context helps in recognition rather than just stating the accuracy.
- **Practical modularity and extensions**
  Our implementation will be modular: the trained model will be packaged so it can later be integrated into an application. This supports the project goal of being extensible to accessibility-focused media applications without claiming to solve the problem of sign language translation.

Conclusion
Koller et al.'s paper [1] is a strong peer-reviewed source for hybrid CNN-HMM techniques for sign language recognition. It shows how sequence-aware probabilistic decoding can significantly improve recognition accuracy compared with techniques that do not adequately account for sequence structure. Koller et al.'s paper is directly relevant to the design and motivation of our final project; meanwhile, our final project is sufficiently unique to be appropriate for a two-person course project within a specific scope, while still suitable for a scientific report due to its experimental design and interpretability/uncertainty-aware nature.

Reference
[1] O. Koller, S. Zargaran, H. Ney, and R. Bowden, "Deep Sign: Hybrid CNN-HMM for Continuous Sign Language Recognition," in *Proc. British Machine Vision Conference (BMVC)*, 2016.